Reflow-Oven — Standard Operating Procedure

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> **Audience:** Maintenance Tech · Line Manager **Applies to:** Line-E, station 3, Reflow-Oven

1. Purpose

The Reflow-Oven melts solder paste to bond SMT components placed by SMT-Placement. A precise multi-zone temperature profile is essential to joint quality. At 45 s ideal cycle, it is one of the longer steps on Line-E.

2. Normal operating envelope

- Ideal cycle time: **45 s**
- Max acceptable cycle time: **55 s**
- Expected reject rate: **≤ 2 %**
- Fault probability per cycle: **0.4 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning here	First check
`Starved`	SMT-Placement (station 2) is not feeding boards.	Verify SMT-Placement state.
`Blocked`	AOI-Inspection (station 4) cannot accept the soldered board.	Verify AOI-Inspection state.

4. Faults & corrective actions

4.1 `Temp-Drift`

- **Symptoms in telemetry:** `fault_type = "Temp-Drift"`. Joint quality rejects rise at AOI; reject types cluster around cold-joint or solder-ball.

- **Likely root cause:** thermocouple drift, zone heater degraded.
- **Corrective action:**
  1. Profile against a Mole / data-logger.
  2. Calibrate or replace thermocouples.
  3. Replace failed heating zone if applicable.
- **Restart criteria:** profile within ± 5 °C of recipe across zones.

4.2 `Conveyor-Slip`

- **Symptoms in telemetry:** `fault_type = "Conveyor-Slip"`. PCBs do not travel through the oven at the right speed; thermal profile broken.
- **Likely root cause:** conveyor belt slippage, drive motor slowing.
- **Corrective action:**
  1. LOTO. Tension the conveyor.
  2. Inspect drive motor; replace if loaded.
- **Restart criteria:** consistent conveyor speed.

4.3 `Element-Fail`

- **Symptoms in telemetry:** `fault_type = "Element-Fail"`. One or more zones fall short of setpoint.
- **Likely root cause:** heating element failed open.
- **Corrective action:**
  1. LOTO. Resistance-check elements; replace the failed zone.
  2. Re-validate profile.
- **Restart criteria:** zone temperatures balanced.

5. Preventive maintenance schedule

- **Daily:** record profile.
- **Weekly:** clean conveyor and tunnel.
- **Monthly:** thermocouple verification.
- **Quarterly:** element check; conveyor service.

6. References

- `Maintenance_Workflow_Overview.pdf`
- `Safety_Lockout_Tagout_Procedure.pdf`
- `Line-E_02_SMT-Placement_SOP.pdf`
- `Line-E_04_AOI-Inspection_SOP.pdf`